

Explaining Unsupervised Acoustic Patterns in Parkinson's Patients' Speech through Fuzzy Linguistic Representations

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Abstract

Speech-based analysis has become an important tool for investigating Parkinson's disease, as vocal alterations reflect complex motor impairments and exhibit substantial variability among speech samples and patients. While unsupervised learning methods are well suited to explore intrinsic acoustic structures without relying on clinical labels, their outputs are often difficult to interpret in medical contexts. This work addresses this limitation by proposing an interpretable unsupervised framework that combines Non-Negative Matrix Factorization with fuzzy linguistic modeling to support human-centered analysis of speech data.

The proposed approach extracts latent acoustic factors from high-dimensional speech features and translates both factor semantics and sample representations into linguistic descriptions grounded in perceptual concepts. This enables the generation of explanations at multiple levels, including latent factors, clusters of samples, and individual recordings, without enforcing class-driven separation. An experimental study on a Parkinson's disease speech dataset shows that the framework uncovers semantically meaningful acoustic dimensions related to voice instability, perturbation, noise, and temporal dynamics. The resulting clusters reflect different patterns of speech variability and show distinctive but non-exclusive relationships with diagnostic status.

The results highlight the potential of fuzzy linguistic representations to enhance the interpretability of unsupervised acoustic models, providing complementary global and local insights into speech variability in Parkinson's disease. The proposed framework supports exploratory analysis and provides a transparent, flexible tool for understanding complex speech patterns in medical applications.

Keywords

Parkinson's disease speech analysis, Unsupervised learning, Non-Negative Matrix Factorization, Fuzzy linguistic modeling, Explainable artificial intelligence

1. Introduction

Acoustic analysis of speech has emerged as a promising source of digital biomarkers for Parkinson's disease (PD), as alterations in voice production are among the earliest and most pervasive symptoms of the disorder [1]. Reduced loudness, monotonic pitch, increased jitter and shimmer, and articulatory imprecision are well-documented manifestations of motor control impairment in PD, making speech a valuable, non-invasive modality for monitoring disease progression and severity [2]. Consequently, machine learning techniques have been widely adopted to analyze acoustic features extracted from patients' voices, supporting tasks such as detection, staging, and longitudinal assessment [3].

While many existing studies focus on supervised learning paradigms for detection or severity estimation, these approaches implicitly assume the availability of reliable, well-aligned clinical labels [4]. In the context of Parkinson's disease speech analysis, this assumption is often difficult to satisfy. Clinical assessments are typically sparse in time, subjective, and not always synchronized with high-frequency acoustic recordings, particularly in real-world monitoring scenarios [5]. Moreover, speech production in Parkinson's disease exhibits substantial intra-patient variability influenced by medication

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cycles, fatigue, and contextual factors, which is not easily captured by discrete clinical labels. As a result, supervised models may overlook meaningful acoustic patterns that do not directly correspond to predefined outcome variables [6]. In this setting, unsupervised learning methods play a crucial role by enabling the exploration of intrinsic data structures without relying on explicit labels. They support the discovery of latent acoustic patterns and variability profiles that are particularly relevant for continuous monitoring and exploratory clinical analysis [7]. However, the outputs of unsupervised models are typically numerical, making their interpretation challenging for clinicians and limiting their practical utility.

Beyond predictive performance, the adoption of machine learning models in medical applications increasingly requires explainability [8]. In clinical contexts, explainable artificial intelligence (XAI) is essential to support trust, accountability, and informed decision-making, enabling clinicians to understand the rationale behind model outputs rather than relying on opaque predictions [9, 10, 11]. This is particularly relevant for Parkinson’s disease, where speech impairments reflect complex and heterogeneous motor and cognitive mechanisms that vary across patients and over time [12, 13, 14]. Motivated by these considerations, we focus on unsupervised learning techniques that can support explainability by design, while remaining compatible with human-centered interpretation in clinical settings.

Non-Negative Matrix Factorization (NMF) is a widely adopted unsupervised learning technique for dimensionality reduction that provides part-based, additive representations of data. Beyond its intrinsic interpretability, NMF is commonly employed for clustering tasks, as the latent representation allows samples to be grouped based on their contributions to the extracted factors, yielding clusters that are directly interpretable in the original feature space [15]. Nevertheless, despite this intrinsic advantage, NMF results are still primarily expressed numerically, which may hinder domain experts’ immediate comprehension and limit their practical usability without additional interpretative mechanisms.

To address this limitation, we adopt a fuzzy logic–based interpretability framework proposed in [16], which enhances NMF by mapping numerical latent representations into linguistic descriptions grounded in human perception. Fuzzy logic provides a natural formalism for modeling gradual concepts and vague boundaries, allowing quantitative values to be expressed through linguistic terms such as low, medium, or high, which are closer to the way domain experts reason about complex phenomena [17, 18]. By translating latent factor contributions and sample representations into fuzzy linguistic variables, the framework bridges the gap between numerical outputs and semantic understanding. This enables the generation of natural language explanations that describe latent factors, clusters, and individual samples in an intuitive yet formally grounded manner. Importantly, fuzzy representations preserve the continuous nature of the underlying data while avoiding rigid thresholds, thus supporting nuanced interpretations of acoustic patterns.

In this work, we apply the proposed framework to acoustic speech recordings collected from subjects with and without Parkinson’s disease [19], with the goal of investigating whether fuzzy linguistic representations can effectively describe latent acoustic structures associated with motor impairment. By analyzing latent factors, clusters, and individual samples, we aim to provide interpretable descriptions of speech variability in Parkinson’s disease, to support exploratory analysis, and to offer insights that may be meaningful to clinicians and researchers. To investigate the use of the proposed framework in an exploratory setting, we formulate the following research questions to address latent factor semantics, cluster-level patterns, and sample-level variability in Parkinson’s disease speech data.

- RQ1** How can fuzzy linguistic representations derived from Non-Negative Matrix Factorization be used to describe the semantics of latent acoustic factors in Parkinson’s disease speech data?
- RQ2** How do fuzzy linguistic explanations support the interpretation of unsupervised acoustic patterns at different levels of analysis, namely latent factors, clusters, and individual samples?
- RQ3** To what extent do the extracted fuzzy-interpretable latent factors capture acoustic variability consistent with clinically relevant characteristics of Parkinson’s disease speech?

The remainder of the paper is organized as follows. Section 2 introduces the proposed interpretability framework based on Non-Negative Matrix Factorization and fuzzy linguistic modeling. Section 3

presents the experimental setting and Section 4 discusses the results obtained from the analysis of Parkinson’s disease speech data. Finally, Section 5 concludes the paper and outlines directions for future work.

2. Materials and Method

2.1. Parkinson’s Disease Speech Dataset

In this work, we use a Parkinson’s disease voice dataset originally introduced in [19]. The dataset consists of biomedical measurements extracted from sustained vowel phonations, a speech task commonly adopted to elicit dysphonic symptoms while minimizing articulatory and linguistic variability. In total, the dataset includes 195 voice recordings collected from 31 subjects, of whom 23 are diagnosed with Parkinson’s disease, and 8 are healthy controls.

Each recording corresponds to a single sustained phonation and is represented by a set of acoustic features designed to capture different aspects of vocal production, including pitch stability, amplitude perturbations, harmonic structure, noise components, and nonlinear dynamics of the speech signal. Although each sample is associated with a binary diagnostic label indicating the presence or absence of Parkinson’s disease, this information is deliberately not used in the present study. Instead, the analysis focuses on exploring latent acoustic patterns and variability across speech samples, while potentially revealing subject-level characteristics, through unsupervised and interpretable modeling.

The acoustic features considered in this study describe different, complementary aspects of voice production that are known to be affected in Parkinson’s disease. For clarity, in Table 1, these features are grouped into five semantic categories, each capturing a distinct dimension of vocal behavior.

Pitch perturbation features describe how stable the pitch of the voice is during sustained phonation. A healthy voice typically exhibits regular vibration of the vocal folds, resulting in a stable pitch over time. Increased pitch perturbation reflects irregular or unstable vocal fold vibration, which may arise from impaired motor control.

Amplitude perturbation features capture variations in voice loudness across consecutive vocal cycles. Rather than measuring average loudness, these features describe how consistently vocal intensity is maintained. Irregular amplitude patterns are often associated with reduced control over vocal effort and breath support.

Harmonic structure features characterize the balance between periodic, harmonic components and aperiodic noise in the speech signal. A clear and stable voice is dominated by harmonic energy, whereas increased noise components reflect breathiness, hoarseness, or turbulence in phonation.

Spectral characteristics summarize global properties of the fundamental frequency, such as its average value and range. These features capture overall pitch level and variability, providing a coarse description of voice register and pitch distribution rather than fine-grained instability.

Finally, *nonlinear dynamics features* model the speech signal as a complex system and capture deviations from regular, predictable behavior. These features are sensitive to subtle irregularities and complex temporal patterns in vocal fold vibration that may not be evident from traditional perturbation measures alone.

Since multiple voice recordings are available for each subject, the analysis is performed at the recording level. Each sustained phonation is treated as an independent speech sample and represented by its own acoustic feature vector. Consequently, the discovered latent factors and clusters describe acoustic patterns at the recording level, although they may indirectly reflect subject-level characteristics.

2.2. Interpretable Unsupervised Learning via Fuzzy Linguistic Modeling

The proposed methodology is designed to support the interpretability of unsupervised acoustic analysis by explicitly modeling the semantic structure of latent representations. Rather than focusing solely on dimensionality reduction, the method provides interpretable abstractions of speech data at multiple levels, enabling domain experts to reason about latent acoustic patterns without relying on labeled data.

Table 1

Acoustic features extracted from sustained vowel phonations. The features capture complementary aspects of voice production affected by Parkinson’s disease, including pitch stability, amplitude perturbations, noise components, and nonlinear vocal dynamics.

Category	Feature	Description
Pitch perturbation	MDVP:Jitter(%)	Relative cycle-to-cycle variability of the fundamental frequency.
	MDVP:Jitter(Abs)	Absolute variation of the fundamental frequency between consecutive cycles.
	MDVP:RAP	Relative average perturbation of pitch over three consecutive cycles.
	MDVP:PPQ	Five-point period perturbation quotient measuring pitch instability.
	Jitter:DDP	Average absolute difference between successive pitch periods.
Amplitude perturbation	MDVP:Shimmer	Relative cycle-to-cycle variability of signal amplitude.
	MDVP:Shimmer(dB)	Amplitude perturbation expressed on a logarithmic scale.
	Shimmer:APQ3	Three-point amplitude perturbation quotient.
	Shimmer:APQ5	Five-point amplitude perturbation quotient.
	MDVP:APQ	Average amplitude perturbation over multiple cycles.
	Shimmer:DDA	Average absolute difference between successive amplitudes.
Harmonic structure	NHR	Noise-to-harmonics ratio, measuring the amount of noise in the voice signal.
	HNR	Harmonics-to-noise ratio, indicating the degree of periodicity in phonation.
Spectral characteristics	MDVP:F0(Hz)	Average fundamental frequency of the speech signal.
	MDVP:Fhi(Hz)	Maximum fundamental frequency observed in the recording.
	MDVP:Flo(Hz)	Minimum fundamental frequency observed in the recording.
Nonlinear dynamics	RPDE	Recurrence period density entropy, capturing irregularities in vocal fold vibration.
	DFA	Detrended fluctuation analysis, measuring long-range temporal correlations.
	PPE	Pitch period entropy, quantifying variability in pitch period sequences.
	Spread1	Nonlinear measure of fundamental frequency variation.
	Spread2	Transformed version of Spread1 emphasizing nonlinearity.
	D2	Correlation dimension estimating the complexity of vocal dynamics.

The approach follows an exploratory paradigm in which latent components, group-level structures, and individual samples are jointly analyzed through linguistic descriptions. This is achieved by combining a low-rank factorization model with fuzzy linguistic modeling, allowing numerical latent representations to be expressed in terms that align with human perception.

2.2.1. Extraction of Latent Acoustic Components and groups of patients

The method adopted in this work, previously proposed in [16], is based on Non-Negative Matrix Factorization, an unsupervised learning technique that both extracts latent acoustic components and groups speech samples according to shared acoustic patterns. NMF is particularly suitable for this task because it produces part-based, additive representations that naturally align with the interpretation of speech signals as combinations of elementary acoustic phenomena.

Let $X \in \mathbb{R}^{n \times m}$ denote the matrix of acoustic features extracted from speech recordings, where rows correspond to acoustic features and columns correspond to speech samples. NMF seeks a low-rank approximation of X by factorizing it into two non-negative matrices $W \in \mathbb{R}^{n \times k}$ and $H \in \mathbb{R}^{k \times m}$, such that:

$$\min_{W, H} \|X - WH\|_F^2 \quad \text{subject to} \quad W \geq 0; H \geq 0, \quad (1)$$

where $k \ll \min(n, m)$ represents the number of latent acoustic components and $\|\cdot\|_F$ denotes the Frobenius norm.

The columns of matrix W encode the latent acoustic components in terms of the contribution of each original acoustic feature. Owing to the non-negativity constraint, each component can be interpreted as an additive combination of acoustic features, facilitating semantic interpretation in the context of Parkinson's disease. In particular, each latent component may capture specific speech characteristics associated with motor impairment, such as increased voice instability or reduced articulatory precision.

Matrix H represents speech samples in the latent acoustic subspace, describing the degree to which each latent component contributes to a given sample. This representation simplifies the original high-dimensional problem and allows meaningful patterns of speech variability to emerge. Sample grouping is then obtained by associating each sample with the latent component that exhibits the highest activation:

$$\text{cluster}(x_j) = \arg \max_{i \in 1, \dots, k} H_{ij}, \quad (2)$$

where x_j denotes the j -th speech sample. As a result, the dataset is partitioned into clusters $\mathcal{C} = \{C_1, C_2, \dots, C_k\}$, where each cluster groups speech samples with similar acoustic profiles driven by dominant latent components.

2.2.2. Linguistic Modeling of Latent Component Semantics

To make latent components interpretable, their composition is modeled using fuzzy linguistic variables, which allow numerical contributions to be expressed in semantic terms. To this end, fuzzy logic is adopted as an extension of Boolean logic that allows partial membership, where elements can belong to a set with a degree in the interval $[0,1]$, rather than through crisp assignments [20]. This enables the association of numerical values with interpretable linguistic terms, such as low, medium, and high. For each acoustic feature, a linguistic domain is defined to represent different degrees of contribution, capturing gradual transitions and the inherent uncertainty of perceptual interpretation.

More formally, let f_r denote the r -th acoustic feature, defined over a numerical domain $\mathcal{U}_r \subset \mathbb{R}$. Each acoustic feature f_r is associated with a fuzzy linguistic variable $L_r = (\mathcal{U}_r; \mathcal{T}_r)$, where $\mathcal{T}_r$ denotes a finite set of linguistic concepts representing different degrees of contribution of the feature. Each linguistic concept $t \in \mathcal{T}_r$ is modeled by a fuzzy set defined over $\mathcal{U}_r$, enabling a gradual and non-discrete characterization of feature relevance.

Given a numerical contribution value $v_r \in \mathcal{U}_r$ associated with acoustic feature f_r , its linguistic characterization is obtained by mapping the numerical value to the most representative linguistic concept:

$$\ell_r = \arg \max_{t \in \mathcal{T}_r} \mu_{r,t}(v_r), \quad (3)$$

where $\ell_r \in \mathcal{T}_r$ denotes the linguistic label assigned to feature f_r , and $\mu_{r,t}$ is the membership function of the fuzzy set associated with linguistic concept t .

This mapping enables the generation of linguistic representations of latent acoustic components, expressing their composition in terms of interpretable acoustic features.

The influence of latent component w_j can be expressed as a linguistic summary over the set of acoustic features:

$$\text{The influence on } w_j \text{ is: } \ell_{1,j} \text{ for } f_1, \ell_{2,j} \text{ for } f_2, \dots, \ell_{n,j} \text{ for } f_n.$$

This representation provides a compact, human-interpretable description of each latent acoustic component, in which numerical feature contributions are mapped to linguistic concepts that reflect their relative influence. Such linguistic profiles facilitate the semantic interpretation of latent structures emerging from acoustic data and support exploratory analysis of speech patterns associated with Parkinson's disease.

2.2.3. Interpretable Representation of Samples in the Latent Space

While latent component semantics describe global acoustic structures, understanding how individual speech samples relate to these components is equally important. Each sample is therefore represented in the latent space as a weighted combination of latent components, reflecting its acoustic composition.

This fuzzy linguistic modeling is also applied to the matrix H , which encodes the activation of each speech sample across the latent acoustic components. Each row of H corresponds to a latent acoustic component, while each column represents a speech sample in terms of its activations.

For each latent acoustic component j , a fuzzy linguistic variable $L_j = (\mathcal{U}_j, \mathcal{T}_j)$ is defined over the numerical domain of activation values $\mathcal{U}_j \subset \mathbb{R}$, where $\mathcal{T}_j$ denotes a set of linguistic concepts describing different degrees of activation of the latent component.

Given a numerical activation value $h_{j,i} \in \mathcal{U}_j$, representing the contribution of latent component j to speech sample i , its linguistic characterization is obtained by assigning the linguistic term that best represents it:

$$\ell_{j,i} = \arg \max_{t \in \mathcal{T}_j} \mu_{j,t}(h_{j,i}), \quad (4)$$

where $\ell_{j,i} \in \mathcal{T}_j$ denotes the linguistic label associated with the activation of component j for sample i .

This mapping enables speech samples and patient groups to be described in linguistic terms with respect to latent acoustic components, thereby supporting interpretable cluster- and sample-level analyses in the latent space. As a result, each speech sample i is described by a linguistic profile over the latent acoustic components, expressing the degree to which each component contributes to the sample, as:

Sample i is characterized by: $\ell_{1,i}$ for w_1 , $\ell_{2,i}$ for w_2 , ..., $\ell_{k,i}$ for w_k .

2.2.4. Explanation of Unsupervised Clusters through Fuzzy Summaries

Similarly, clusters of samples can be represented linguistically. To this end, a representative prototype is defined for each cluster by computing the average activation of its samples in the latent acoustic space.

The representative vector of cluster C_i is defined as:

$$\bar{\mathbf{h}}^{(i)} = \frac{1}{|C_i|} \sum_{p \in C_i} \mathbf{h}_p, \quad (5)$$

where $\mathbf{h}_p = (h_{1,p}, \dots, h_{k,p})^\top$ denotes the latent representation of sample p .

Each component $\bar{h}_{ij}$ of the cluster representative is then mapped to its corresponding linguistic label using the fuzzy linguistic variables defined for the latent acoustic components: $\ell_{ij} = \arg \max_{t \in \mathcal{T}_j} \mu_{j,t}(\bar{h}_{ij})$.

As a result, each cluster can be described through a linguistic description of the latent acoustic components:

Samples in C_i are generally characterized by: $\ell_{1,i}$ for w_1 , $\ell_{2,i}$ for w_2 , ..., $\ell_{k,i}$ for w_k .

This representation provides an interpretable, high-level characterization of clusters of speech samples, describing common activation patterns of latent acoustic components and supporting semantic analysis of speech variability in Parkinson's disease.

3. Experimental settings

The experimental study aims to assess the ability of the proposed framework to uncover meaningful acoustic structures and to produce human-interpretable explanations of speech patterns associated with Parkinson's disease.

As a first step, Non-Negative Matrix Factorization was applied to determine the appropriate number k of latent acoustic components to extract from the data.

Subsequently, fuzzy linguistic variables and the corresponding fuzzy sets were defined to support the mapping of numerical latent representations into interpretable linguistic concepts. Finally, the resulting fuzzy-based explanations were generated and analyzed, providing linguistic descriptions of latent acoustic components, clusters of speech samples, and individual samples, which will be discussed in the following section.

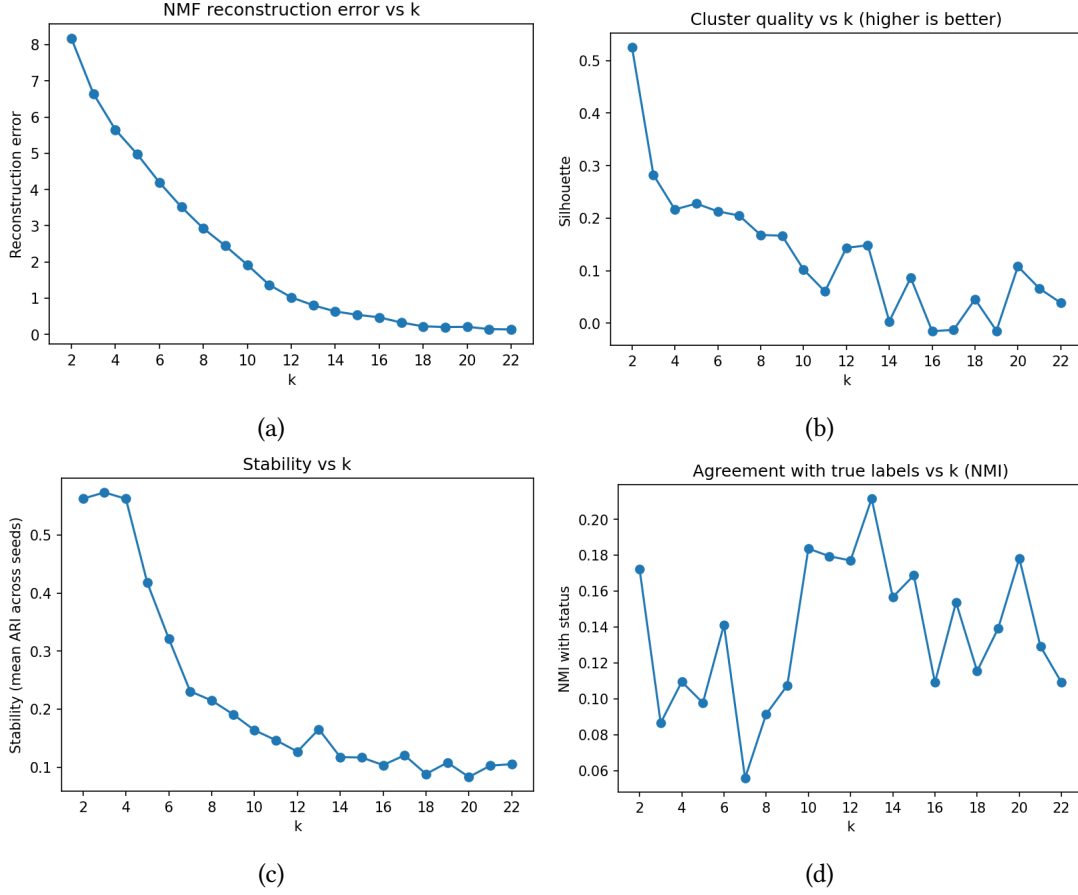


Figure 1: Multi-criteria evaluation for selecting the number of latent factors k in NMF. (a) Reconstruction error as a function of k , showing decreasing gains with increasing model complexity. (b) Cluster quality in the latent space measured by the Silhouette coefficient. (c) Clustering stability across random initializations quantified by the mean Adjusted Rand Index (ARI). (d) Agreement between the unsupervised clustering and the available health-status labels measured by Normalized Mutual Information (NMI).

All experiments were implemented in Python¹. The NMF algorithm and data preprocessing steps were carried out using the scikit-learn library², while fuzzy linguistic modeling and the generation of explanations were implemented using the scikit-fuzzy library³.

3.1. NMF Parameter Configuration

For the NMF algorithm, the coordinate descent optimization strategy was adopted, together with an SVD-based initialization, as these choices have been shown to provide effective convergence and stable factorization results in high-dimensional settings [15].

The number of latent factors k to be extracted was determined within a multi-criteria experimental setting that integrates complementary evaluation measures, including reconstruction error analysis, cluster quality assessment, stability analysis across multiple random initializations, and agreement with known class labels (see Fig. 1). This integrated evaluation strategy ensures that the selected value of k yields not only an accurate numerical approximation of the original data but also robust, stable, and interpretable latent representations, which are essential for the subsequent fuzzy linguistic analysis.

Specifically, reconstruction accuracy was quantified by analyzing the Frobenius-norm reconstruction error between the input data matrix and its low-rank NMF approximation [21]. As shown in Fig. 1(a),

¹Python: <https://www.python.org/>

²scikit-learn: <https://scikit-learn.org/0.19/modules/generated/sklearn.decomposition.NMF.html>

³scikit-fuzzy: <https://pythonhosted.org/scikit-fuzzy/>

the reconstruction error decreases rapidly for small values of k , while the rate of improvement markedly slows down beyond $k \approx 5$, indicating diminishing returns in terms of numerical approximation for higher model complexity.

Cluster quality was evaluated in the latent space using the Silhouette coefficient [22]. As illustrated in Fig. 1(b), the Silhouette score attains its highest values for low values of k , with a noticeable drop when increasing the number of latent factors. In particular, values beyond $k = 5$ do not improve cluster separability and are associated with increasingly fragmented cluster structures.

To account for the non-convex nature of the NMF optimization problem, clustering stability was assessed by repeating the factorization under multiple random initializations and measuring the consistency of the resulting cluster assignments using the Adjusted Rand Index (ARI) [23]. Fig. 1(c) shows that stability is highest for small values of k and progressively deteriorates as k increases, with a pronounced decline after $k = 5$, indicating reduced robustness of the latent representations for more complex factorizations. It is important to note that health-status labels are never used during NMF training, latent factor extraction, cluster assignment, or fuzzy explanation generation. The labels are considered only a posteriori, as an external validation criterion, to assess whether the discovered unsupervised structures are related to the known diagnostic categories.

Finally, although the proposed framework is inherently unsupervised, agreement between the induced clusters and the known health status labels was examined as a complementary external validation criterion using Normalized Mutual Information (NMI) [24]. As reported in Fig. 1(d), NMI values exhibit local maxima for moderate values of k , with $k = 5$ providing a favorable trade-off between alignment with clinical labels and model simplicity, without using label information during training.

Considering all criteria jointly, the number of latent factors was set to $k = 5$, with k explored in the range [2, 23]. This choice reflects a balanced compromise between reconstruction accuracy, cluster separability, stability across initializations, external clinical agreement, and semantic interpretability. Although lower values of k yielded slightly higher Silhouette and stability scores, they produced a less expressive latent representation and tended to merge acoustic phenomena that were separated when using five factors. Conversely, larger values of k improved reconstruction accuracy only marginally, while reducing stability and interpretability. Therefore, $k = 5$ was considered the most appropriate configuration, as it provides a moderate-dimensional representation that remains robust and interpretable, while facilitating the emergence of meaningful latent acoustic patterns for subsequent fuzzy linguistic modeling and the generation of coherent and informative explanations.

3.2. Fuzzy Linguistic Modeling Configuration

Another important aspect of the proposed framework concerns the definition of the fuzzy sets used for linguistic interpretation. To obtain linguistic interpretations of numerical features, fuzzy sets were constructed for each selected attribute. Since the variables considered differ in scale, units of measurement, and statistical distributions, defining a single global fuzzy partition would be meaningless. Therefore, a separate universe of discourse was defined for each feature based on its minimum and maximum observed values, after normalization to the interval [0, 1].

For each feature, three linguistic terms were considered, namely *low*, *medium*, and *high*, providing a coarse yet interpretable linguistic granulation of the feature domain. These linguistic terms were modeled using Gaussian membership functions, chosen for their smoothness and their ability to represent gradual transitions between contribution levels. In particular, a fully defined Gaussian function was used for the central (*medium*) term, while open-ended Gaussian functions were adopted for the lower (*low*) and upper (*high*) terms, allowing extreme values to be represented without imposing hard saturation boundaries.

The parameters of the fuzzy sets were determined in a data-driven manner using empirical quantiles of the normalized feature distribution, as this approach allows the linguistic partitions to adapt to the underlying data distribution of each feature while remaining robust to scale differences and outliers. Specifically, the 33rd percentile q_{33} was used to characterize the lower region of the data, the median q_{50} represented the central tendency, and the 67th percentile q_{67} marked the transition toward higher

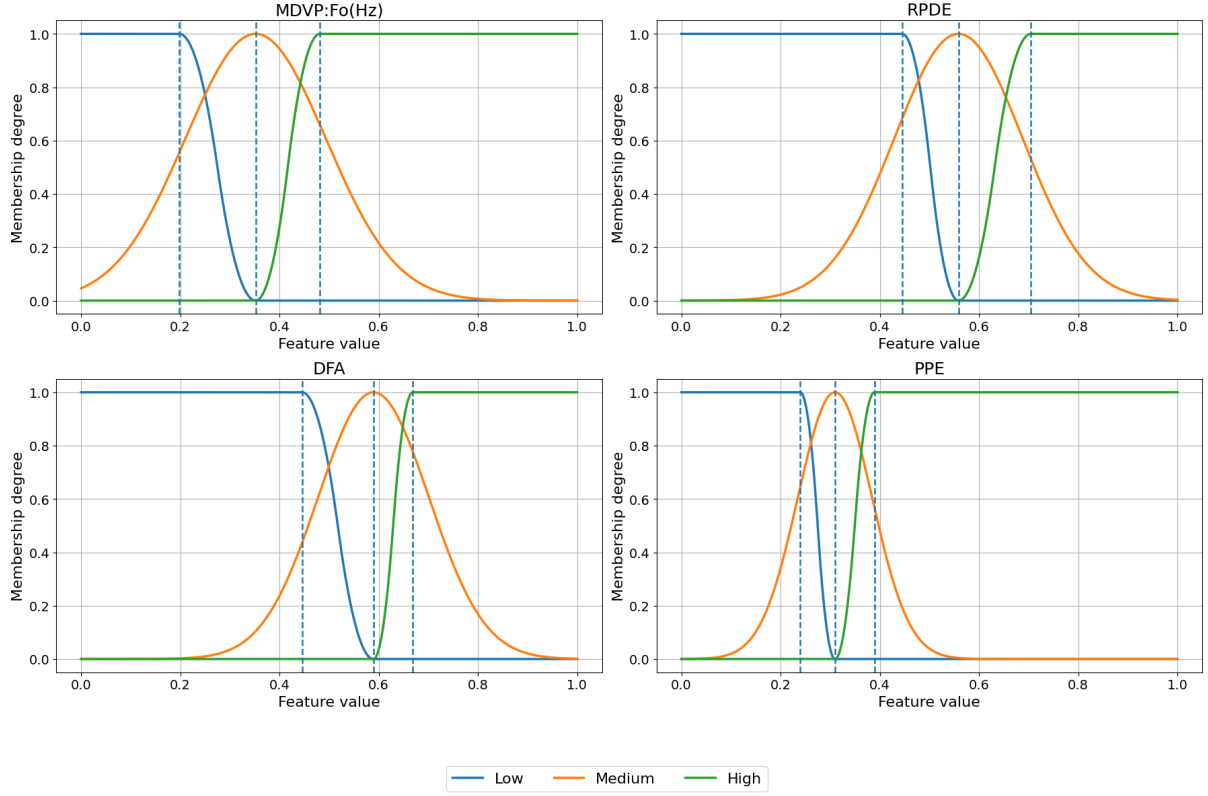


Figure 2: Fuzzy sets (Low, Medium, High) for selected normalized features, defined using empirical quantiles and heterogeneous membership function shapes.

values.

Formally, let $x \in [0, 1]$ denote a normalized feature value and let q_{33} , q_{50} , and q_{67} be the empirical 33rd, 50th, and 67th percentiles computed from its distribution. The membership function associated with the linguistic term low was defined using a Z-shaped function as follows:

$$\mu_{\text{Low}}(x) = \begin{cases} 1, & x \leq q_{33}, \\ 1 - 2 \left(\frac{x - q_{33}}{q_{50} - q_{33}} \right)^2, & q_{33} < x \leq \frac{q_{33} + q_{50}}{2}, \\ 2 \left(\frac{x - q_{50}}{q_{50} - q_{33}} \right)^2, & \frac{q_{33} + q_{50}}{2} < x < q_{50}, \\ 0, & x \geq q_{50}. \end{cases} \quad (6)$$

The medium fuzzy set was modeled using a Gaussian membership function centered at the median:

$$\mu_{\text{Medium}}(x) = \exp \left(-\frac{(x - q_{50})^2}{2\sigma^2} \right), \quad (7)$$

where the standard deviation was set to

$$\sigma = \frac{q_{67} - q_{33}}{2}, \quad (8)$$

Finally, the linguistic term high was defined using an S-shaped membership function:

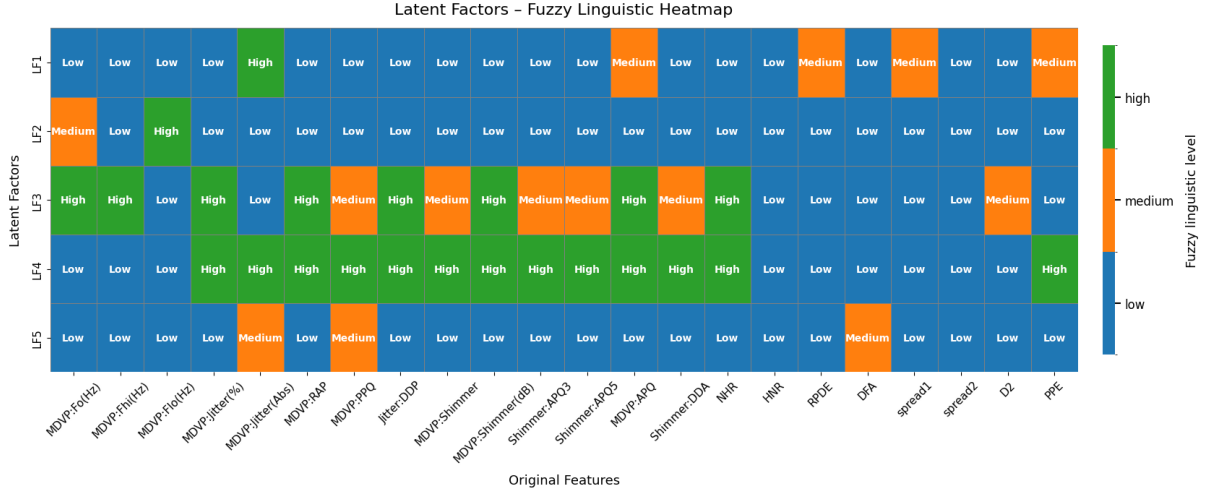


Figure 3: Linguistic contribution (Low, Medium, High) of each original feature to the five latent factors extracted by NMF. Rows correspond to latent factors (LF1–LF5), while columns represent the original acoustic features. The heatmap provides an interpretable summary of the latent structure, highlighting which features predominantly influence each latent factor and supporting the semantic analysis of the learned representation.

$$\mu_{\text{High}}(x) = \begin{cases} 0, & x \leq q_{50}, \\ 2 \left(\frac{x - q_{50}}{q_{67} - q_{50}} \right)^2, & q_{50} < x \leq \frac{q_{50} + q_{67}}{2}, \\ 1 - 2 \left(\frac{x - q_{67}}{q_{67} - q_{50}} \right)^2, & \frac{q_{50} + q_{67}}{2} < x < q_{67}, \\ 1, & x \geq q_{67}. \end{cases} \quad (9)$$

To provide an intuitive illustration of the adopted fuzzy partitions, exemplary fuzzy sets for selected acoustic features are illustrated in Fig. 2. Specifically, four fuzzy linguistic variables are shown, corresponding to the attributes MDVP:F0, RPDE, DFA, and PPE. For each feature, the fuzzy sets associated with the linguistic terms *low*, *medium*, and *high* are depicted in blue, orange, and green, respectively. As can be observed, the shapes and relative positions of the fuzzy sets vary across features, reflecting differences in their empirical data distributions and highlighting the data-driven nature of the proposed fuzzy modeling.

4. Results

This section presents the experimental results obtained with the proposed NMF–fuzzy framework. The analysis is conducted at multiple levels in order to provide both global and local interpretations of the learned latent representation. At a global level, the latent factors extracted by Non-negative Matrix Factorization are analyzed to characterize the dominant acoustic patterns captured in the data and their role in shaping the induced clustering structure. At the local level, fuzzy linguistic modeling is used to generate interpretable descriptions of individual speech samples, enabling fine-grained, sample-level explanations. In both cases, latent factors are described in terms of their associated acoustic features, providing a semantic interpretation that bridges global cluster-level insights and local sample-level explanations.

The original data were transformed from the initial 22-dimensional feature space into a 5-dimensional latent space defined by the latent factors LF1, LF2, LF3, LF4, and LF5. This dimensionality reduction provides a compact and interpretable representation of the data, in which each latent factor captures a distinct underlying acoustic pattern. Based on the latent factor activations, five clusters c^1 , c^2 , c^3 , c^4 , and c^5 were identified in the reduced space, each corresponding to a characteristic latent component.

4.1. Semantic Interpretation of Latent Acoustic Factors

To facilitate a clearer understanding of the proposed methodology, before presenting and analyzing the generated explanations, we first inspect an intermediate representation in which the numerical values of the NMF basis matrix are mapped to their corresponding fuzzy linguistic labels. Fig. 3 reports the fuzzy linguistic heatmap of the NMF basis matrix, providing a compact and interpretable view of how the original acoustic features contribute to each latent factor. Each cell encodes the linguistic activation level (*low*, *medium*, *high*) of a given feature's contribution to a specific latent component, providing a global overview of the semantic structure learned by the model. For consistency and interpretability, the same color mapping used to define the fuzzy sets is applied to the heatmap.

Although each acoustic latent factor is obtained as a linear combination of multiple features in the original space, the heatmap reveals that, for each factor, only a subset of acoustic features exhibits a stronger influence. To derive a concise and interpretable representation of the latent factors, the analysis therefore focuses on features showing *medium* and *high* linguistic activation levels, which are considered the most informative contributors. Features associated with *low* activation levels are omitted for clarity, as they provide negligible semantic contribution and may introduce visual or conceptual noise.

Based on this criterion, the fuzzy representations of the acoustic latent factors can be summarised by the dominant original features that influence each component, as discussed below.

The influence on **LF1** is:

High for MDVP:Jitter(Abs),
Medium for MDVP:APQ,
Medium for RPDE,
Medium for spread1,
Medium for PPE.

The influence on **LF2** is:

Medium for MDVP:F0(Hz),
High for MDVP:F1(Hz).

The influence on **LF3** is:

High for MDVP:F0(Hz),
High for MDVP:F1(Hz),
High for MDVP:Jitter(%),
High for MDVP:RAP,
Medium for MDVP:PPQ,
High for Jitter:DDP,
Medium for MDVP:Shimmer,
High for MDVP:Shimmer(dB),
Medium for Shimmer:APQ3,
Medium for Shimmer:APQ5,
High for Shimmer:APQ,
Medium for Shimmer:DDA,
High for NHR,
Medium for D2.

The influence on **LF4** is:

High for MDVP:Jitter(%),
High for MDVP:Jitter(Abs),
High for MDVP:RAP,
High for MDVP:PPQ,
High for Jitter:DDP,
High for MDVP:Shimmer,
High for MDVP:Shimmer(dB),
High for Shimmer:APQ3,
High for Shimmer:APQ5,
High for MDVP:APQ,
High for Shimmer:DDA,
High for NHR,
High for PPE.

The influence on **LF5** is:

Medium for MDVP:Jitter(Abs),
Medium for MDVP:PPQ,
Medium for DFA.

The acoustic Latent Factor LF1 is primarily characterized by high absolute pitch perturbation (MDVP:Jitter(Abs)) and medium contributions from amplitude perturbation (MDVP:APQ), nonlinear dynamics (RPDE, Spread1), and pitch instability (PPE). This combination suggests a latent pattern associated with irregular vocal fold vibration and reduced phonatory stability, reflecting subtle motor control impairments. Increased jitter and PPE values are commonly associated with micro-instabilities in vocal fold oscillation, while elevated RPDE and nonlinear measures indicate greater complexity and

unpredictability in voice dynamics, phenomena frequently observed in Parkinson’s disease speech, even at early stages.

The acoustic Latent Factor LF2 is dominated by spectral characteristics, with a medium contribution from the average fundamental frequency (MDVP:F₀) and a high contribution from the minimum fundamental frequency (MDVP:F_{lo}). This factor appears to capture baseline pitch characteristics and pitch floor variations, which may be related to reduced pitch range and monotonicity, a well-documented hallmark of hypokinetic dysarthria in Parkinson’s disease. Lower or constrained fundamental frequency values can reflect reduced laryngeal flexibility and diminished prosodic modulation.

The acoustic Latent Factor LF3 exhibits strong contributions from both pitch perturbation (jitter measures, RAP, PPQ, DDP) and amplitude perturbation (shimmer-related features), along with harmonic noise components (NHR) and nonlinear dynamics (D2). This factor represents a global voice degradation pattern, in which instability in both the frequency and amplitude domains co-occurs with increased noise and reduced harmonic structure. Such a profile indicates impaired neuromuscular control of phonation, resulting in breathy, rough, and unstable voice quality.

The acoustic Latent Factor LF4 is characterized by consistently high contributions across nearly all pitch and amplitude perturbation measures, as well as high NHR and PPE. This factor captures an extreme pattern of severe phonatory instability, combining pronounced frequency perturbations, amplitude fluctuations, increased noise components, and high pitch period entropy. From a clinical perspective, this latent factor may be associated with advanced motor impairment that affects voice production, in which both the regularity and efficiency of vocal fold vibration are substantially compromised.

The acoustic Latent Factor LF5 is mainly influenced by medium contributions from pitch perturbation (MDVP:Jitter(Abs), MDVP:PPQ) and long-range temporal correlation (DFA). This factor appears to encode temporal correlation and self-similarity properties of the speech signal, reflecting alterations in the temporal organization of phonation rather than purely local perturbations. Changes in DFA values have been associated with altered motor control strategies and reduced adaptability of the speech production system in Parkinson’s disease.

Overall, these latent factors provide a multidimensional and interpretable decomposition of Parkinson’s disease speech, with each factor highlighting a distinct acoustic dimension, ranging from pitch and amplitude instability to spectral degradation and nonlinear temporal dynamics. The fuzzy linguistic representation allows these complex patterns to be expressed in human-understandable terms, supporting both global and local interpretation of speech variability.

4.2. Global Explanations at the Cluster Level

In a manner analogous to the analysis conducted for latent acoustic factors, fuzzy linguistic explanations were derived at the cluster level by aggregating the latent-factor activations of samples in the same group. This process yields representative linguistic profiles that summarise the dominant latent patterns characterizing each cluster.

The resulting cluster-level explanations reveal a clear and interpretable structure in which each group is primarily driven by a single latent factor, while the remaining factors exhibit consistently low activation. Rather than indicating a trivial clustering outcome, this behavior suggests that the proposed NMF–fuzzy framework effectively disentangles distinct acoustic dimensions, leading to semantically homogeneous clusters that can be interpreted along specific axes of speech impairment.

Samples in **cluster c⁽¹⁾** are generally characterized by
Medium values of LF1 AND Low values of LF2 AND Low values of LF3 AND Low values of LF4 AND
Low values of LF5

Samples in **cluster c⁽²⁾** are generally characterized by
Low values of LF1 AND Medium values of LF2 AND Low values of LF3 AND Low values of LF4 AND
Low values of LF5

Samples in **cluster c⁽³⁾** are generally characterized by

Low values of $LF1$ AND Low values of $LF2$ AND Medium values of $LF3$ AND Low values of $LF4$ AND Low values of $LF5$

Samples in **cluster $c^{(4)}$** are generally characterized by Low values of $LF1$ AND Low values of $LF2$ AND Low values of $LF3$ AND Medium values of $LF4$ AND Low values of $LF5$

Samples in **cluster $c^{(5)}$** are generally characterized by Low values of $LF1$ AND Low values of $LF2$ AND Low values of $LF3$ AND Low values of $LF4$ AND Medium values of $LF5$

Specifically, cluster $c^{(1)}$ is primarily characterized by a predominance of $LF1$, reflecting irregular vocal fold vibration and moderate phonatory instability driven by pitch perturbation and nonlinear dynamics. This cluster captures speech patterns associated with subtle alterations in motor control, in which instability is present but not accompanied by pronounced spectral or amplitude degradation.

In contrast, cluster $c^{(2)}$ emerges as mainly governed by $LF2$, highlighting baseline pitch characteristics and pitch floor variations. This separation indicates that fundamental frequency-related properties constitute an independent dimension of variability in Parkinson’s disease speech, distinct from perturbation- and noise-related phenomena.

Cluster $c^{(3)}$ is dominated by $LF3$, which encodes a combination of pitch and amplitude instability together with increased noise components. Samples in this group exhibit a global degradation of voice quality, consistent with reduced neuromuscular control of phonation and the emergence of roughness or breathiness.

While, cluster $c^{(4)}$ is primarily influenced by $LF4$. The dominance of this factor reflects extensive phonatory instability across multiple acoustic dimensions, combining strong frequency and amplitude perturbations with elevated noise and entropy measures. This cluster represents speech patterns where both regularity and efficiency of vocal fold vibration are substantially compromised.

Finally, cluster $c^{(5)}$ is characterized by $LF5$, which captures alterations in temporal correlation and long-range dependencies in the speech signal. Unlike the previous groups, this cluster reflects changes in the temporal organization of phonation rather than local perturbations, suggesting alternative or compensatory motor control strategies affecting speech production.

Overall, the cluster-level explanations indicate that the learned representation organizes samples along well-separated and semantically meaningful acoustic dimensions. The emergence of clusters dominated by individual latent factors facilitates interpretation by allowing each group to be associated with a specific aspect of speech variability in Parkinson’s disease, thereby supporting exploratory analysis and clinically informed interpretation.

4.3. Local Explanations at the Sample Level

In addition to cluster-level descriptions, the proposed framework also enables the generation of sample-level explanations. Thanks to the low-rank representation, each observation is mapped from the original high-dimensional feature space into a compact latent space, where relevant patterns become more tractable. Within this representation, each sample is characterized by the relative activation of the five latent factors, whose semantic meaning has been previously established. This allows clinicians and researchers to inspect individual recordings through their latent factor profiles. To illustrate the interpretability of the proposed framework, four representative samples were selected from different regions of the latent space. The selected examples include both samples that closely match their cluster prototype and those exhibiting more atypical activation profiles, highlighting the framework’s ability to support both prototypical and individual-level interpretations.

The following explanations illustrate the latent factor profiles of four representative samples.

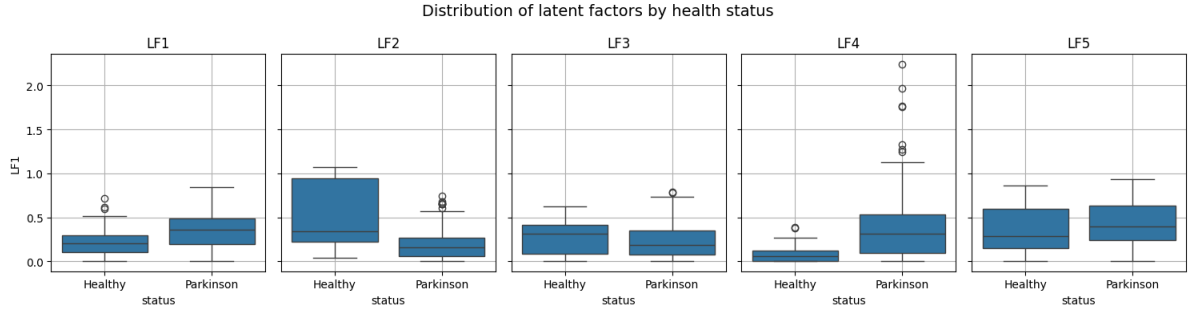
The influence on **sample 138** is :

Low for $LF1$, Low for $LF2$, Low for $LF3$, Medium for $LF4$, Low for $LF5$

Table 2

Distribution of Healthy and Parkinson’s disease samples across clusters.

Cluster	Healthy	Parkinson
Cluster 0	0.135	0.865
Cluster 1	0.700	0.300
Cluster 2	0.524	0.476
Cluster 3	0.000	1.000
Cluster 4	0.153	0.847

**Figure 4:** Distribution of latent factor activations for Healthy and Parkinson’s disease samples, shown separately for each latent factor.

The influence on **sample 16** is :

Medium for LF1, Low for LF2, Low for LF3, Low for LF4, Low for LF5

The influence on **sample 155** is :

Medium for LF1, Low for LF2, Low for LF3, Low for LF4, Low for LF5

The influence on **sample 96** is :

Low for LF1, Low for LF2, Low for LF3, Low for LF4, Low for LF5

Taken together, global and local explanations provide complementary perspectives on the data. Cluster-level descriptions offer a coherent and synthetic view of the dominant latent patterns characterizing a cluster of speech samples, whereas sample-level explanations capture finer-grained variations that may deviate from the cluster prototype. These local differences highlight sample-specific peculiarities that may be relevant in clinical settings, where detailed inspection of individual recordings can provide additional insights beyond group-level patterns. As a result, the proposed approach supports flexible interpretability across different levels of granularity, allowing practitioners to switch between a global understanding of the data distribution and a local analysis of individual cases, depending on the analytical and clinical needs.

4.4. Cluster Composition with Respect to Clinical Classes

The NMF-based analysis is designed to explain latent acoustic structures using automatically extracted acoustic latent features, providing an interpretable representation of the data. Rather than focusing on correlations with predefined class labels, the objective is to uncover latent relationships among speech samples and latent associations among acoustic features, which may exist independently of diagnostic categories that are not always known or explicitly defined.

Nevertheless, since the dataset is labeled and the application domain is medical, it is natural to investigate how the identified clusters relate to the original classes. The following analysis is therefore intended as an exploratory study, aimed at discussing the positioning of the discovered clusters with respect to the initial class labels, without altering the unsupervised nature of the proposed approach.

Table 2 reports the distribution of Healthy and Parkinson’s disease samples across the five clusters identified by the NMF-based analysis. As expected in an unsupervised setting, clusters do not directly correspond to the original diagnostic classes. However, a non-uniform distribution of classes emerges.

Clusters 0, 3, and 4 are largely dominated by recordings associated with Parkinson’s disease, with cluster 3 exclusively composed of pathological samples, whereas cluster 1 mainly includes recordings from healthy subjects. Cluster 2 presents a nearly balanced composition, yet relates to health status through distincts.

This distribution indicates that the clustering process captures latent acoustic structures that are not class-driven but nevertheless relate to the health status through different activation profiles. This interpretation is further supported by Fig. 4, which shows that the distributions of latent factor activations differ between Healthy and Parkinson samples, even when the same latent dimensions are involved.

By jointly considering cluster-level descriptions and the acoustic semantics of the latent factors, it emerges that clusters dominated by Parkinson’s patients tend to exhibit stronger activations of latent dimensions associated with voice perturbation, instability, and noise-related acoustic features. These include measures related to jitter, shimmer, and noise-to-harmonics ratios, which are widely recognized as markers of dysphonic and pathological speech. In contrast, clusters with a higher proportion of healthy subjects are associated with more moderate or lower activations of these perturbation-related latent components, suggesting more stable and regular phonatory patterns.

The mixed composition observed in cluster 2 further reinforces this view. This cluster aggregates samples with intermediate latent-factor activations, indicating that certain acoustic patterns are transversal across health conditions and may manifest along a continuum rather than as strictly class-specific signatures. In this sense, Parkinson’s disease appears not as a single homogeneous acoustic profile, but as a condition characterized by different combinations and intensities of latent acoustic traits.

Overall, this cross-analysis suggests that the latent factors extracted by NMF encode acoustic dimensions consistent with clinically relevant characteristics and that cut across diagnostic labels. While these dimensions are not explicitly optimized for class separation, their differential expression across Healthy and Parkinson’s subjects provides valuable insights into how pathological speech characteristics emerge from underlying acoustic patterns. This further confirms the suitability of the proposed approach for exploratory analysis in medical contexts, where uncovering latent structures and their semantic interpretation is often more informative than enforcing rigid class boundaries.

5. Conclusion

In this paper, we investigated how unsupervised acoustic patterns in Parkinson’s disease speech data can be made interpretable through fuzzy linguistic representations. By combining Non-Negative Matrix Factorization with fuzzy logic-based modeling, we proposed a human-centered framework that provides explanations at multiple levels of analysis, including latent acoustic factors, sample clusters, and individual speech recordings. The approach was validated on a well-known Parkinson’s disease speech dataset, illustrating its ability to uncover meaningful latent structures while preserving interpretability.

The study was guided by three research questions, which we address below.

RQ1: *How can fuzzy linguistic representations derived from Non-Negative Matrix Factorization be used to describe the semantics of latent acoustic factors in Parkinson’s disease speech data?* The results show that fuzzy linguistic modeling enables a clear and semantically grounded interpretation of the latent acoustic factors extracted by NMF. By mapping numerical feature contributions to linguistic terms, each latent factor can be characterized by dominant acoustic characteristics, such as pitch perturbation, amplitude instability, noise components, and nonlinear dynamics.

RQ2: *How do fuzzy linguistic explanations support the interpretation of unsupervised acoustic patterns at different levels of analysis, namely latent factors, clusters, and individual samples?* The proposed framework supports interpretability across multiple levels of granularity. Latent factors provide a global view of the underlying acoustic dimensions, cluster-level descriptions capture shared patterns among groups of samples, and sample-level explanations highlight individual variability. These complementary perspectives enable a flexible and comprehensive understanding of the data.

RQ3: *To what extent do the extracted fuzzy-interpretable latent factors capture acoustic variability*

consistent with clinically relevant characteristics of Parkinson's disease speech? The analysis suggests that the extracted latent factors capture acoustic variability consistent with clinically relevant characteristics of Parkinson's disease speech. Although the model is not trained using class labels, the distribution of latent factor activations differs across Healthy and Parkinson samples, indicating that the learned acoustic dimensions reflect relevant speech-related patterns rather than arbitrary structures.

Despite the promising results, some limitations of this study should be acknowledged. The analysis is conducted on a relatively small dataset, which may limit the generalisability of the findings. In addition, the evaluation is primarily qualitative and interpretive, lacking a formal user study with domain experts to assess the effectiveness and usability of the generated explanations. Furthermore, the proposed analysis focuses on NMF, an interpretable, prototype-based factorization method, combined with fuzzy linguistic modeling. Although this choice is coherent with the objective of generating semantically meaningful latent acoustic factors, alternative prototype-based or unsupervised representation learning methods could also be investigated.

These limitations open several directions for future research. First, the framework could be extended to longitudinal speech data to analyze how latent acoustic factors and their linguistic descriptions evolve over time [25], potentially supporting the monitoring of disease progression. Second, future studies could compare different prototype-based or unsupervised methods and analyze the stability of the generated explanations across alternative latent representations. Such an analysis would not aim to identify a single universally optimal model, since different methods may capture different aspects of the acoustic space, but rather to assess how robust and complementary the resulting explanations are. Third, integrating additional speech tasks or multimodal data sources (e.g., articulatory or prosodic features) could enrich the latent representations and broaden their clinical relevance. Moreover, integrating large language models to enhance the explanations generated by the proposed framework represents a promising research direction [26]. Finally, involving clinicians in evaluating the generated explanations would provide valuable feedback on their usability and interpretability in real-world diagnostic and monitoring scenarios [27].

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Declaration on Generative AI

During the preparation of this work, the authors used GPT-5.3 (ChatGPT, OpenAI) and Grammarly exclusively for language refinement, including grammar correction and improvement of the English writing style. No AI tools were used for generating scientific content, ideas, or interpretations. All the content of the manuscript, including the methodology, analysis, and conclusions, was developed by the authors. The authors reviewed and edited all outputs and take full responsibility for the content of this publication.

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